My Toolbox

Eight tools for starting a problem

AGES
10-13



1 Spot the pattern

Find the rule, test it on a new case, then use it to jump far.



2 List every way

Count every case once with a systematic order; rule out repeats.



3 Draw it

Sketch and label a model; let the picture show what stays the same.



4 Make a table

Tabulate cases, find the change in each column, extend or generalise.



5 Work backwards

Reverse the operations from the result, then confirm by running forwards.



6 Guess and check

Guess, check, adjust using how far off you were; stop when every condition holds.



7 Try something simpler

Solve 1, 2, 3 cases first, find how the answer grows, then handle the real size.



8 Act it out

Model or simulate the situation (objects, fold, cut, dice) and record the outcomes.

Look back

Look at one problem from today

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Pick one problem from today. There are no right answers. You can pass. Nobody reads it out.

1 **Did I answer the question that was asked (with units or a reason)?**

☐ yes ☐ not yet

2 **Is my answer reasonable? (Estimate or compare.)**

☐ yes ☐ not yet

3 **How can I check it another way? (Go backwards, estimate or try a smaller case.)**

I checked by ...

.....

.....

My tool today

_____

My name _____

☐ I am still stuck (my tutor will come) ☐ pass